Pearson Edexcel Level 1/Level 2 – style practice paper

GCSE (9–1) Mathematics

1MA1/1H & 2H – Higher Tier

Topic 1: Number

Centre Number		**Candidate Number**	
Surname		**Other names**	

Practice questions in the style of Pearson Edexcel GCSE Mathematics (1MA1). Every question is original; the layout, command words, mark allocations and mark scheme (M, A, B marks) follow the real exam format. Calculator rules are given in the instructions below.

Time allowed	About 105 minutes (roughly 1 minute per mark)
Instructions	• Use **black** ink or ball-point pen. • Answer **all** questions. • Answer the questions in the spaces provided – there may be more space than you need. • Calculators must **not** be used in Section A. Calculators may be used in Section B. • Diagrams are **NOT** accurately drawn, unless otherwise indicated. • You must **show all your working**. • **Unless stated, give non-exact answers to 3 significant figures.**
Information	• The total mark for this paper is **107**. • The marks for **each** question are shown in brackets – use this as a guide as to how much time to spend on each question. • Questions labelled with an asterisk (*) are ones where the quality of your written communication will be assessed.
Advice	• Read each question carefully before you start to answer it. • Keep an eye on the time. • Try to answer every question. • Check your answers if you have time at the end.

Name: _____ Date: _____ Mark: _____ / 107

Answer ALL questions.

Write your answers in the spaces provided. You must write down all the stages in your working.

Section A: Non-calculator

You must not use a calculator for questions 1 to 16.

1 Work out 4.6 × 0.35 **(2)**

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(Total for Question 1 is 2 marks)

2 Work out $2\frac{1}{3} + 1\frac{3}{4}$. Give your answer as a mixed number. **(3)**

.....

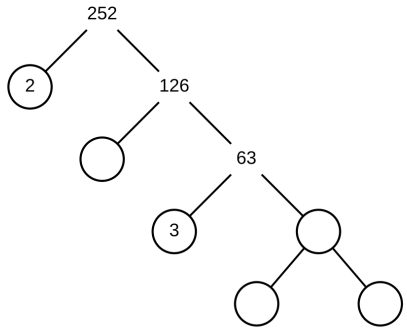
(Total for Question 2 is 3 marks)

3 Work out $2\frac{2}{5} \div 1\frac{1}{5}$ **(3)**

.....

(Total for Question 3 is 3 marks)

4 Here is a factor tree for 252



(a) Complete the factor tree. Write a number in each of the four empty circles. **(2)**

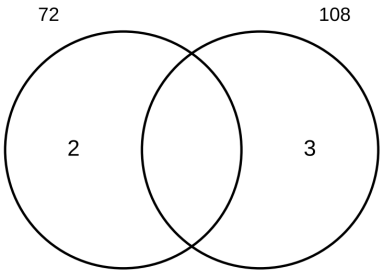
(b) Write 252 as a product of its prime factors. **(2)**

.....

(Total for Question 4 is 4 marks)

5 $72 = 2 \times 2 \times 2 \times 3 \times 3$ and $108 = 2 \times 2 \times 3 \times 3 \times 3$

(a) Complete the Venn diagram so that the prime factors of 72 and 108 are in the correct regions. One number in each of the two outer regions has been done for you. **(2)**



(b) Use your Venn diagram to find the Highest Common Factor (HCF) of 72 and 108. **(1)**

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(c) Use your Venn diagram to find the Lowest Common Multiple (LCM) of 72 and 108. **(2)**

.....

(Total for Question 5 is 5 marks)

6 Two buses leave a bus station at 8:00 am. One bus leaves every 12 minutes. The other bus leaves every 18 minutes. At what time will both buses next leave the station together? **(3)**

.....

(Total for Question 6 is 3 marks)

7 Write these numbers in order of size. Start with the smallest. $0.6 \quad \frac{5}{8} \quad 61\% \quad \frac{7}{12}$ **(2)**

.....

(Total for Question 7 is 2 marks)

8 Sara had some money. She spent $\frac{2}{5}$ of her money on a coat. She spent $\frac{1}{3}$ of the money she had left on shoes. She then had £48. How much money did Sara have to start with? **(4)**

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(Total for Question 8 is 4 marks)

9 Show that $0.\dot{4}\dot{5} = \frac{5}{11}$ **(2)**

.....

(Total for Question 9 is 2 marks)

10 Write $0.2\dot{1}$ as a fraction in its simplest form. **(3)**

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(Total for Question 10 is 3 marks)

11 (a) Work out the value of 5^{-2} **(1)**

.....

(b) Work out the value of $64^{\frac{2}{3}}$ **(2)**

.....

(c) Work out the value of $\left(\frac{27}{8}\right)^{-\frac{2}{3}}$ **(3)**

.....

(d) Simplify fully $\frac{(2x^3y)^4}{8x^5y^2}$ **(3)**

.....

(Total for Question 11 is 9 marks)

12 (a) Write 0.00372 in standard form. **(1)**

.....

(b) Work out $(3 \times 10^5) \times (4 \times 10^{-2})$. Give your answer in standard form. **(2)**

.....

(c) Work out $\frac{6.4 \times 10^7}{1.6 \times 10^{-3}}$. Give your answer in standard form. **(2)**

.....

(d) Work out $3.2 \times 10^6 + 4.5 \times 10^5$. Give your answer in standard form. **(2)**

.....

(Total for Question 12 is 7 marks)

13 (a) Simplify $\sqrt{50} + \sqrt{18}$. Give your answer in the form $a\sqrt{b}$ where a and b are integers. **(2)**

.....

(b) Expand and simplify $(3 + \sqrt{2})(4 - \sqrt{2})$ **(2)**

.....

(c) Rationalise the denominator of $\frac{6}{\sqrt{3}}$ **(2)**

.....

(d) Show that $\frac{10}{3-\sqrt{5}}$ can be written in the form $\frac{a+b\sqrt{5}}{2}$ where a and b are integers. **(3)**

Find the values of a and b.

.....

(Total for Question 13 is 9 marks)

14 Work out $1.44 \div 0.012$ **(2)**

.....

(Total for Question 14 is 2 marks)

15 A restaurant menu has 4 starters, 6 main courses and 3 desserts.

(a) Hari chooses one starter, one main course and one dessert. How many different meals can he choose? **(2)**

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(b) A four-digit PIN uses the digits 0 to 9. No digit can be used more than once. How many different PINs are possible? **(2)**

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(Total for Question 15 is 4 marks)

16 (a) Work out an estimate for $\frac{4.82 \times 19.7}{0.52}$ **(3)**

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(b) Is your estimate in part (a) bigger than or smaller than the exact answer? Give a reason for your answer. **(2)**

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.....

(Total for Question 16 is 5 marks)

Section B: Calculator

A calculator may be used for questions 17 to 27.

17 Rita invests £6000 for 3 years in a savings account. The account pays compound interest at 2.5% per year. Work out the value of her investment at the end of 3 years. **(3)**

£ .....

(Total for Question 17 is 3 marks)

18 The price of a house increased by 24% to £310 000. Work out the price of the house before the increase. **(3)**

£ .....

(Total for Question 18 is 3 marks)

19 In a sale, all prices are reduced by 15%. The sale price of a coat is £212.50. Work out the normal price of the coat. **(3)**

£ .....

(Total for Question 19 is 3 marks)

20 Ali bought a motorbike for £4800. He sold it for £3900. Work out his percentage loss. Give your answer correct to 1 decimal place. **(3)**

.....

(Total for Question 20 is 3 marks)

21 A new car costs £14 500. Its value falls by 15% each year.

(a) Work out the value of the car after 3 years. **(3)**

£ .....

(b) After how many whole years will the value of the car first be less than £7000? You must show your working. **(2)**

.....

(Total for Question 21 is 5 marks)

22 A population of bacteria in a dish is 500. It increases by 12% every hour.

Work out the number of bacteria after 6 hours. Give your answer to the nearest whole number. **(3)**

.....

(Total for Question 22 is 3 marks)

23 Bank A pays simple interest at 3.2% per year. Bank B pays compound interest at 3% per year. Jamal has £2000 to invest for 5 years.

Which bank pays more interest over the 5 years? You must show how you get your answer. **(4)**

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(Total for Question 23 is 4 marks)

24 The length of a rectangle is 8.6 cm, correct to 1 decimal place. The width of the rectangle is 4.3 cm, correct to 1 decimal place.

(a) Write down the error interval for the length. **(2)**

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(b) Work out the upper bound for the area of the rectangle. **(2)**

.....

(c) Work out the lower bound for the perimeter of the rectangle. **(2)**

.....

(Total for Question 24 is 6 marks)

25 Ben runs 120 metres, correct to the nearest 10 metres. His time is 14.0 seconds, correct to 3 significant figures.

Work out the upper bound for Ben's average speed. Give your answer to 3 significant figures. **(4)**

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(Total for Question 25 is 4 marks)

26 The mass of the Earth is 5.97×10^{24} kg. The mass of Jupiter is 1.9×10^{27} kg.

Work out how many times greater the mass of Jupiter is than the mass of the Earth. Give your answer in standard form correct to 3 significant figures. **(3)**

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(Total for Question 26 is 3 marks)

27 Work out $\dfrac{3.8^2 - \sqrt{14.5}}{2.6 + 1.35}$. Write down all the figures on your calculator display. Then give your answer correct to 3 significant figures. **(3)**

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(Total for Question 27 is 3 marks)

TOTAL FOR PAPER IS 107 MARKS

END

MARK SCHEME – Topic 1: Number

Guidance on marking: M marks are method marks, awarded for a correct method (even if the final answer is wrong). **A** marks are accuracy marks, awarded only if the relevant M mark has been gained. **B** marks are independent marks (no method needed). **C** marks are communication marks. **P** marks are process marks (a correct approach to a problem). **ft** = follow through from a previous incorrect answer. **oe** = or equivalent. **cao** = correct answer only. **SC** = special case. If the answer on the answer line differs from the working, the answer line is marked unless the working is clearly the intended answer. Correct answers with no working score full marks unless the question says 'You must show your working'.

Question	Working	Answer	Mark	Notes
1	M1 46 × 35 = 1610 (or 4.6 × 35 = 161 or 0.46 × 3.5 = 1.61 shown) A1 1.61	1.61	2 AO1	Condone 1.610
2	M1 $\frac{7}{3}+\frac{7}{4}$ or $\frac{28}{12}+\frac{21}{12}$ or $3+\frac{4}{12}+\frac{9}{12}$ M1 $\frac{49}{12}$ or $3\frac{13}{12}$ A1 $4\frac{1}{12}$	$4\frac{1}{12}$	3 AO1	
3	M1 $\frac{12}{5}\div\frac{6}{5}$ M1 $\frac{12}{5}\times\frac{5}{6}$ or $\frac{12}{6}$ A1 2	2	3 AO1	
4(a)	B2 all four correct: 2, 21, 3, 7 in the circles (B1 for any two correct)	2, 21, 3, 7	2 AO1	
4(b)	M1 2 × 2 × 3 × 3 × 7 (all primes, product seen) A1 $2^2\times3^2\times7$	$2^2\times3^2\times7$	2 AO1	
5(a)	B2 for 2, 2, 3, 3 in the overlap and nothing else added (B1 for two correct numbers in the overlap, or all four with one extra)	Overlap: 2, 2, 3, 3	2 AO1	
5(b)	B1 36	36	1 AO1	
5(c)	M1 $2^3\times3^3$ or 8 × 27 or all five numbers multiplied A1 216	216	2 AO1	
6	M1 attempt to find the LCM of 12 and 18 (e.g. lists of multiples) A1 36 A1 8:36 am (oe)	8:36 am	3 AO3	Allow 08:36 or 36 minutes after 8
7	M1 for converting to a common form, e.g. 0.6, 0.625, 0.61, 0.58(3…) A1 $\frac{7}{12}$, 0.6, 61%, $\frac{5}{8}$	$\frac{7}{12}$, 0.6, 61%, $\frac{5}{8}$	2 AO1	
8	M1 $1-\frac{2}{5}=\frac{3}{5}$ (money left after the coat) M1 $\frac{2}{3}\times\frac{3}{5}=\frac{2}{5}$ (fraction left at end) M1 $\frac{2}{5}$ of start = 48, so $48\div2\times5$ A1 120	£ 120	4 AO3	

Question	Working	Answer	Mark	Notes
9	M1 $x = 0.454545\ldots$, $100x = 45.4545\ldots$ C1 $99x = 45$ so $x = \frac{45}{99} = \frac{5}{11}$		2 AO2	
10	M1 $10x = 2.111\ldots$ and $100x = 21.111\ldots$ M1 $90x = 19$ A1 $\frac{19}{90}$	$\frac{19}{90}$	3 AO1	
11(a)	B1 $\frac{1}{25}$ (oe, 0.04)	$\frac{1}{25}$	1 AO1	
11(b)	M1 $\sqrt[3]{64} = 4$ A1 16	16	2 AO1	
11(c)	M1 $\left(\frac{8}{27}\right)^{\frac{2}{3}}$ M1 $\left(\frac{2}{3}\right)^2$ A1 $\frac{4}{9}$	$\frac{4}{9}$	3 AO1	
11(d)	M1 $(2x^3y)^4 = 16x^{12}y^4$ M1 for $16 \div 8 = 2$ or x^{12-5} or y^{4-2} A1 $2x^7y^2$	$2x^7y^2$	3 AO1	
12(a)	B1 3.72×10^{-3}	3.72×10^{-3}	1 AO1	
12(b)	M1 12×10^3 A1 1.2×10^4	1.2×10^4	2 AO1	
12(c)	M1 4×10^{10} or $6.4 \div 1.6 = 4$ and $10^{7-(-3)}$ A1 4×10^{10}	4×10^{10}	2 AO1	
12(d)	M1 $3\,200\,000 + 450\,000$ or $32 \times 10^5 + 4.5 \times 10^5$ A1 3.65×10^6	3.65×10^6	2 AO1	
13(a)	M1 $5\sqrt{2}$ or $3\sqrt{2}$ seen A1 $8\sqrt{2}$	$8\sqrt{2}$	2 AO1	
13(b)	M1 at least 3 terms of $12 - 3\sqrt{2} + 4\sqrt{2} - 2$ correct A1 $10 + \sqrt{2}$	$10 + \sqrt{2}$	2 AO1	
13(c)	M1 $\frac{6\sqrt{3}}{3}$ A1 $2\sqrt{3}$	$2\sqrt{3}$	2 AO1	
13(d)	M1 $\frac{10(3+\sqrt{5})}{(3-\sqrt{5})(3+\sqrt{5})}$ M1 denominator $= 9 - 5 = 4$ A1 $\frac{30+10\sqrt{5}}{4} = \frac{15+5\sqrt{5}}{2}$, so $a = 15$, $b = 5$	a = 15, b = 5	3 AO2	
14	M1 $1440 \div 12$ or $144 \div 1.2$ or $1.44 \div 0.012 = 1440 \div 12$ A1 120	120	2 AO1	
15(a)	M1 $4 \times 6 \times 3$ A1 72	72	2 AO1	

Question	Working	Answer	Mark	Notes
15(b)	M1 $10 \times 9 \times 8 \times 7$ A1 5040	5040	2 AO3	
16(a)	M1 rounds to 1 s.f.: 5, 20, 0.5 M1 $\frac{5 \times 20}{0.5}$ A1 200	200	3 AO1	
16(b)	M1 identifies that 5 and 20 round up and 0.5 rounds down, or that 4.82 and 19.7 both round up C1 bigger than, because the numerator is rounded up and the denominator is rounded down (oe)		2 AO2	
17	M1 6000×1.025^3 (oe) M1 (dep) 6000×1.076890625 A1 £6461.34 (accept 6461.343…)	£ 6461.34	3 AO1	
18	M1 $310000 \div 1.24$ or $1.24 \times x = 310000$ M1 (dep) $310000 \div 1.24$ evaluated or $x = \frac{310000}{124} \times 100$ A1 250 000	£ 250 000	3 AO1	
19	M1 $100 - 15 = 85$ or 0.85 M1 $212.50 \div 0.85$ A1 250	£ 250	3 AO1	
20	M1 $4800 - 3900 = 900$ M1 $\frac{900}{4800} \times 100$ A1 18.8	18.8 %	3 AO1	
21(a)	M1 14500×0.85^3 (oe) M1 14500×0.614125 A1 8904.81 (accept 8904.8125)	£ 8904.81	3 AO1	
21(b)	M1 attempt at 14500×0.85^n for $n = 4$ and 5 (values 7569… and 6433…) A1 5 (years)	5 years	2 AO3	
22	M1 500×1.12^6 M1 $500 \times 1.973822…$ A1 987	987	3 AO1	
23	M1 simple: $2000 \times 0.032 \times 5$ (= 320) M1 compound: 2000×1.03^5 (= 2318.55…) A1 total in A = 2320 and total in B = 2318.55 (or interest 320 and 318.55) C1 Bank A (pays £1.45 more)	Bank A	4 AO3	
24(a)	B1 8.55 B1 $8.55 \leq L < 8.65$ with correct inequality signs	8.55 ≤ L < 8.65	2 AO1	B1 for each correct bound; condone 8.65 as upper bound
24(b)	M1 8.65×4.35 A1 37.6275 (accept 37.63)	37.6275 cm²	2 AO1	
24(c)	M1 $2 \times (8.55 + 4.25)$ A1 25.6	25.6 cm	2 AO1	

Question	Working	Answer	Mark	Notes
25	B1 125 (upper bound of distance) B1 13.95 (lower bound of time) M1 $125 \div 13.95$ A1 8.96	8.96 m/s	4 AO3	
26	M1 $\frac{1.9 \times 10^{27}}{5.97 \times 10^{24}}$ M1 318.25… A1 3.18×10^{2}	3.18×10^{2}	3 AO1	
27	M1 $14.44 - 3.80789\ldots$ or numerator 10.6321… M1 $\frac{10.632\ldots}{3.95}$ A1 2.69 (calculator display 2.6916…)	2.69	3 AO1	

Where marks are lost in Topic 1

Use this page after marking. Tick the ones that cost you marks, then redo those questions closed-book.

- ☐ **Calculator slips:** on Section B always write the unrounded value first (e.g. 6461.34375) and only round on the final line. Rounding early lost marks on the compound-interest questions.
- ☐ **Compound interest:** use a single multiplier ($\times 1.025^3$), not 2.5% × 3. A 15% fall is ×0.85, not ×0.15.
- ☐ **Reverse percentages:** divide by the multiplier (÷1.24 or ÷0.85). Never take 24% off the new price.
- ☐ **Recurring decimals:** subtract two multiples of x that have the same recurring tail. Always finish by simplifying the fraction.
- ☐ **Surds:** simplify each surd first ($\sqrt{50} = 5\sqrt{2}$). When rationalising a two-term denominator multiply top and bottom by the conjugate.
- ☐ **Bounds:** for a÷b the upper bound is UB(a) ÷ LB(b). Write which bounds you are using.
- ☐ **Standard form:** after multiplying or dividing, check the front number is between 1 and 10 and adjust the power.
- ☐ **Fractions:** in 'fraction of the remainder' questions work out what is *left* after each step and use the fraction of the original.